STAT 153 & 248 - Time Series
Lecture Fifteen

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1 Recap: Periodogram

Consider a time series dataset $y_0, \ldots, y_{n-1}$. One of most common tools in time data analysis is the periodogram. It is given by:

$$I(j/n) := \frac{|b_j|^2}{n} \qquad \text{for } 0 < \frac{j}{n} \leq \frac{1}{2}$$

where b_j is the Discrete Fourier Transform (DFT) defined by

$$b_j := \sum_{t=0}^{n-1} y_t \exp\left(-\frac{2\pi i j t}{n}\right).$$

For simplicity, we shall sometimes assume (in the sequel) that n is odd and $m = (n-1)/2$. Then the periodogram ordinate is defined for $j = 1, \ldots, m$.

The periodogram shows which frequency components contribute most strongly to the behaviour of the observed time series. It is useful for identifying the dominant cyclical patterns or periodicities within a dataset. Indeed, we saw previously (Lectures 8-9) that (provided $0 < j/n < 1/2$)

$$RSS(j/n) := \min_{\beta_0,\beta_1,\beta_2} \sum_t \left(y_t - \beta_0 - \beta_1 \cos(2\pi j t/n) - \beta_2 \sin(2\pi j t/n)\right)^2 = \sum_t (y_t - \bar{y})^2 - 2I(j/n),$$

which shows that the periodogram is directly related of how well the sinusoid at frequency j/n fits the observed data.

More generally if $f_1, \ldots, f_k$ are all Fourier frequencies strictly lying between 0 and 1/2, then (as discussed in Lecture 9)

$$RSS(f_1, \ldots, f_k) = \min_{\beta_0,\beta_{j1},\beta_{j2}, 1 \leq j \leq k} \sum_t \left(y_t - \beta_0 - \sum_{j=1}^{k} \left[\beta_{j1} \cos(2\pi f_j t) + \beta_{j2} \sin(2\pi f_j t)\right]\right)^2$$

can be written in terms of the periodogram as:

$$RSS(f_1, \ldots, f_k) = \sum_t (y_t - \bar{y})^2 - 2\sum_{j=1}^{k} I(f_j).$$

If n is odd and if we use all the Fourier frequencies j/n to fit the data, then we would get a perfect fit leading to zero RSS (because there will be as many coefficient parameters as the data points in that linear regression model). The above formula then gives:

$$\sum_t (y_t - \bar{y})^2 = 2 \sum_{j=1}^{(n-1)/2} I(j/n) \qquad \text{if } n \text{ is odd.} \tag{1}$$

This shows that the periodogram $I(j/n)$ provides a decomposition of the sample variance. More precisely, we can view $2I(j/n)$ as the portion of sample variance that is explained by sines and cosines at frequency j/n.

When n is even, the formula needs to be modified as:

$$\sum_t (y_t - \bar{y})^2 = 2 \sum_{j=1}^{(n/2)-1} I\left(\frac{j}{n}\right) + I\left(\frac{1}{2}\right) \qquad \text{if } n \text{ is even.}$$

2 Smoothing the periodogram

For most real datasets, the periodogram appears rough and noisy, especially when looking at its plot on the log-scale (i.e., when looking at the plot of log-periodogram). In such cases, researchers attempt to smooth the periodogram to get a clean summary of the trend in the periodogram without getting distracted by noise. What is a good way of smoothing the periodogram?

A nice way of smoothing is via the use of an appropriate model. Previously, we saw for estimating a smooth trend function, say μ_t, from observed time series data y_t, a natural model is:

$$y_t = \mu_t + \epsilon_t \qquad \text{with } \epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2).$$

We can try to use a similar model for the periodogram. Modeling $I(j/n)$ as a smooth trend function plus Gaussian noise does not make much sense for multiple reasons: (a) the periodogram is a positive quantity so using the normal distribution may not be a good idea, (b) the noise in the periodogram is more pronounced on the log-scale so we should probably use an additive noise model for the log-periodogram as opposed to the periodogram directly. In other words, we should use a multiplicative noise model for the periodogram.

The model that we shall use here is:

$$I(j/n) = f(j/n)\eta_j \qquad \text{for } j = 1, \ldots, m.$$

where $f(j/n)$ is a smooth function that represents the trend in the periodogram, and

$$\eta_j \overset{\text{i.i.d}}{\sim} \frac{1}{2}\chi^2_2.$$

χ^2_2 is the chi-squared distribution with 2 degrees of freedom; its density is given by:

$$f_{\chi^2_2}(x) := \frac{1}{2}\exp(-x/2) I\{x > 0\}.$$

The density of $\chi^2_2/2$ is then given by:

$$f_{\chi^2_2/2}(x) = 2 f_{\chi^2_2}(2x) = \exp(-x) I\{x > 0\}.$$

The above is the density of the Standard Exponential Distribution (the exponential distribution $Exp(\lambda)$ has density $\lambda e^{-\lambda x} I\{x > 0\}$; the above corresponds to $\lambda = 1$). Thus

$$\frac{\chi^2_2}{2} = Exp(1).$$

Our model can therefore also be written as:

$$I(j/n) = f(j/n)\eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} Exp(1). \tag{2}$$

This is a multiplicative noise model. We can rewrite it in additive form by taking logarithms:

$$\log I(j/n) = \log f(j/n) + \log \eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} Exp(1). \tag{3}$$

The noise in this additive representation is captured by the terms $\log \eta_j, 1 \le j \le n$. These variables do not have mean zero (from the internet $\mathbb{E} \log \eta_j \approx -0.5772$). Therefore $\log f(j/n)$ does the represent the mean of $\log I(j/n)$ (in fact, it is strictly larger than the mean of $\log I(j/n)$). Rather, $f(j/n)$ represents the mean of $I(j/n)$.

We shall refer to (2) or (3) as the Spectrum Model.

3 Power Spectral Density

Since the mean of the $Exp(1)$ distribution equals 1, the mean of $I(j/n)$ in the Spectrum Model (2) is given by $f(j/n)$. This quantity is known as the **power** of frequency j/n:

$$f(j/n) := \text{power of frequency } j/n = \text{mean of } I(j/n) \text{ in model (2)}.$$

If we plot the points $(j/n, f(j/n))$ for $j = 1, \ldots, m$ and join the neighboring points by lines, we get a continuous function plot. This function is known as the **power spectral density** and is defined on $[0, 0.5]$.

This definition of the power spectral density is not rigorous. For a rigorous treatment, see any book on time series (e.g., Chapter 4 of Shumway and Stoffer; or the book "Spectral Analysis for Univariate Time Series" by Percival and Walden).

Note that the power spectral density is not a probability density function in the sense that it does not integrate to one.

4 Spectrum Model from DFT

In (2) or (3), we formulated the spectrum model in terms of the periodogram or log-periodogram. We can also describe it in terms of the DFT. Specifically, in terms of the DFT $b_0, \ldots, b_{n-1}$ of the data $y_0, \ldots, y_{n-1}$, the model is given by:

$$\text{Re}(b_j), \text{Im}(b_j) \overset{\text{i.i.d}}{\sim} N(0, \gamma_j^2) \tag{4}$$

for $j = 1, \ldots, m$ where $m = (n-1)/2$ (we are assuming that n is odd). The unknown parameters in this model are $\gamma_1^2, \ldots, \gamma_m^2$ and γ_j represents the strength of sinusoids at frequency j/n.

By definition of the periodogram, we get

$$I(j/n) := \frac{|b_j|^2}{n} = \frac{1}{n}\left(\mathrm{Re}(b_j)^2 + \mathrm{Im}(b_j)^2\right) = \frac{2\gamma_j^2}{n}\frac{1}{2}\left(\left(\frac{\mathrm{Re}(b_j)}{\gamma_j}\right)^2 + \left(\frac{\mathrm{Im}(b_j)}{\gamma_j}\right)^2\right).$$

By the assumption (4), we get

$$\eta_j := \left(\left(\frac{\mathrm{Re}(b_j)}{\gamma_j}\right)^2 + \left(\frac{\mathrm{Im}(b_j)}{\gamma_j}\right)^2\right) \overset{\text{i.i.d}}{\sim} \chi_2^2.$$

Therefore the model (4) implies that

$$I(j/n) = \frac{2\gamma_j^2}{n}\eta_j \qquad \text{with } \eta_j \overset{\text{i.i.d}}{\sim} \frac{\chi_2^2}{2} = Exp(1).$$

The is exactly the same as (2) with the identification:

$$f(j/n) = \frac{2\gamma_j^2}{n}.$$

Because of this equivalence, we shall treat (4) as another formulation of the spectrum model.

5 Rewriting the Model in terms of y_t

The model (4) is written in terms of the DFT b_j. Because the original data can be written in terms of the DFT, we can convert (4) into a specification for the original data y_t. This will lead us to another model representation which is directly in terms of the original data y_t.

The key to this is the following formula which shows how to write the real-valued data y_t in terms of its DFT. We shall assume n is odd (and $m = (n-1)/2$) for simplicity. With a simple modification, this formula also holds for n even.

$$y_t = \frac{b_0}{n} + \sum_{j=1}^{m}\left(\frac{2\mathrm{Re}(b_j)}{n}\cos\left(\frac{2\pi jt}{n}\right) + \frac{-2\mathrm{Im}(b_j)}{n}\sin\left(\frac{2\pi jt}{n}\right)\right). \tag{5}$$

In other words, when n is odd and $m = (n-1)/2$, we have

$$y_t = \beta_0 + \sum_{j=1}^{m}\left(\beta_{1j}\cos\frac{2\pi jt}{n} + \beta_{2j}\sin\frac{2\pi jt}{n}\right) \tag{6}$$

where, for $j = 1, \ldots, m$,

$$\beta_0 = \frac{b_0}{n} \quad \beta_{1j} = \frac{2\mathrm{Re}(b_j)}{n} \quad \beta_{2j} = -\frac{2\mathrm{Im}(b_j)}{n} \tag{7}$$

The formula (6) holds for every dataset $y_0, \ldots, y_{n-1}$. As a result, the model (4) is equivalent to (6) with

$$\beta_{1j} = \frac{2\mathrm{Re}(b_j)}{n} \sim N\left(0, \frac{4}{n^2}\gamma_j^2\right) \quad \text{and} \quad \beta_{2j} = \frac{-2\mathrm{Re}(b_j)}{n} \sim N\left(0, \frac{4}{n^2}\gamma_j^2\right).$$

The spectrum model therefore has the following three equivalent definitions:

- **Definition 1**: $\text{Re}(b_1), \text{Im}(b_1), \ldots, \text{Re}(b_m), \text{Im}(b_m)$ are all independent with $\text{Re}(b_j), \text{Im}(b_j) \overset{\text{i.i.d}}{\sim} N(0, \gamma_j^2)$ for $j = 1, \ldots, m$.
- **Definition 2**:

$$y_t = \beta_0 + \sum_{j=1}^{m} \left(\beta_{1j} \cos \frac{2\pi j t}{n} + \beta_{2j} \sin \frac{2\pi j t}{n} \right) \tag{8}$$

with $\beta_{11}, \beta_{21}, \beta_{12}, \beta_{22}, \ldots, \beta_{1m}, \beta_{2m}$ all independent with

$$\beta_{1j}, \beta_{2j} \overset{\text{i.i.d}}{\sim} N(0, \tau_j^2).$$

- **Definition 3**: $I(j/n) = f(j/n)\eta_j$ with $\eta_j \overset{\text{i.i.d}}{\sim} Exp(1)$.

These two definitions are equivalent because of (6) and (7). The three sets of parameters $(\gamma_1^2, \ldots, \gamma_m^2)$ and $(\tau_1^2, \ldots, \tau_m^2)$ and $(f(1/n), \ldots, f(m/n))$ are related via:

$$\frac{4\gamma_j^2}{n^2} = \tau_j^2 \iff \frac{2\gamma_j}{n} = \tau_j$$

which is because

$$\text{Re}(b_j) \sim N(0, \gamma_j^2) \implies \frac{2\text{Re}(b_j)}{n} \sim N\left(0, \frac{4\gamma_j^2}{n^2}\right),$$

and

$$f(j/n) = \frac{2\gamma_j^2}{n} = \frac{n\tau_j^2}{2} \qquad \text{for } j = 1, \ldots, m.$$

6 Regularized Estimation

Parameter estimation is done by maximizing likelihood with a regularization penalty which ensures smoothness of the estimates (without the regularization penalty, we would overfit in the sense that the estimate of $f(j/n)$ would coincide with the periodogram $I(j/n)$).

To obtain the likelihood, we can use any of the three definitions of the spectrum model. If we use Definition 1 (i.e., (4)), the likelihood will be given by:

$$\begin{aligned}
&\prod_{j=1}^{m} \frac{1}{\gamma_j} \exp\left(-\frac{(\text{Re}(b_j))^2}{2\gamma_j^2}\right) \frac{1}{\gamma_j} \exp\left(-\frac{(\text{Im}(b_j))^2}{2\gamma_j^2}\right) \\
&= \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{(\text{Re}(b_j))^2 + (\text{Im}(b_j))^2}{2\gamma_j^2}\right) = \prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{|b_j|^2}{2\gamma_j^2}\right).
\end{aligned}$$

We can rewrite the above likelihood in terms of the periodogram (because $I(j/n) = |b_j|^2/n$) as follows:

$$\prod_{j=1}^{m} \frac{1}{\gamma_j^2} \exp\left(-\frac{nI(j/n)}{2\gamma_j^2}\right). \tag{9}$$

We can also use the definition (2) and write the likelihood directly using the exponential distribution. Note that the density of η_j is $\exp(-x)I\{x > 0\}$ and the density of $f(j/n)\eta_j$ is $\frac{1}{f(j/n)} \exp(-\frac{x}{f(j/n)})I\{x > 0\}$. Thus the likelihood (joint density of $I(j/n), 1 \le j \le m$) is:

$$\prod_{j=1}^{n} \frac{1}{f(j/n)} \exp\left(-\frac{I(j/n)}{f(j/n)}\right). \tag{10}$$

The above is equivalent to (9) (up to proportionality) because $f(j/n) = 2\gamma_j^2/n$.

The negative log-likelihood corresponding to (9) is

$$\sum_{j=1}^{m} \left(2 \log \gamma_j + \frac{nI(j/n)}{2\gamma_j^2} \right).$$

For optimization purposes we work with the logarithms of γ_j. Let $\alpha_j = \log \gamma_j$. The negative log-likelihood in terms of α_j is

$$\sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j} \right).$$

If we directly minimize the above with respect to α_j (without any additional regularization), we get

$$\alpha_j = \log \sqrt{\frac{nI(j/n)}{2}} \quad \text{and} \quad \gamma_j^2 = e^{2\alpha_j} = \frac{nI(j/n)}{2}.$$

This basically means that the γ_j^2 parameters fully interpolate the periodogram leading to full overfitting. For more meaningful estimation, we need to add regularization. If we assume smoothness of α_j, we can add the penalty $\sum_{j=2}^{m-1}((\alpha_{j+1} - \alpha_j) - (\alpha_j - \alpha_{j-1}))^2$ or $\sum_{j=2}^{m-1} |(\alpha_{j+1} - \alpha_j) - (\alpha_j - \alpha_{j-1})|$ to the negative log-likelihood. This leads to the estimators $\hat{\alpha}_t^{\text{ridge}}(\lambda)$ and $\hat{\alpha}_t^{\text{lasso}}(\lambda)$ which are defined as the minimizers of

$$\sum_{j=1}^{m} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j} \right) + \lambda \sum_{j=2}^{m-1} ((\alpha_{j+1} - \alpha_j) - (\alpha_j - \alpha_{j-1}))^2$$

and

$$\sum_{t=1}^{n} \left(2\alpha_j + \frac{nI(j/n)}{2} e^{-2\alpha_j} \right) + \lambda \sum_{j=2}^{m-1} |(\alpha_{j+1} - \alpha_j) - (\alpha_j - \alpha_{j-1})|$$

respectively. The penalties encourage smoothness in $\{\alpha_j\}$, leading to more stable and interpretable estimates for $\{\gamma_j^2\}$.

Once we obtain estimates $\hat{\alpha}_j$ of α_j, we can convert them to estimates of γ_j via $\hat{\gamma}_j = \exp(\hat{\alpha}_j)$ and then to estimates of $f(j/n)$ via $\hat{f}(j/n) = 2\hat{\gamma}_j^2/n$.

7 Proof of the identity (5)

We will assume n is odd and $m = (n-1)/2$.

In Lecture 9, we defined the following. For a Fourier frequency j/n strictly lying between 0 and 1/2 (i.e., $0 < j/n < 1/2$),

$$\mathbf{c}^j := (1, \cos(2\pi j/n), \cos(2\pi 2j/n), \ldots, \cos(2\pi(n-1)j/n))^T$$

and

$$\mathbf{s}^j := (0, \sin(2\pi j/n), \sin(2\pi 2j/n), \ldots, \sin(2\pi(n-1)j/n))^T.$$

These are the vectors obtained by evaulating $\cos(2\pi ft)$ and $\sin(2\pi ft)$ with Fourier Frequency $f = j/n$ at time points $t = 0, 1, \ldots, (n-1)$.

We also mentioned in Lecture 9 that

$$\mathbf{c}^0, \mathbf{c}^1, \mathbf{s}^1, \ldots, \mathbf{c}^{\frac{n-1}{2}}, \mathbf{s}^{\frac{n-1}{2}}. \tag{11}$$

are orthogonal vectors in $\mathbb{R}^n$.

Because n orthogonal vectors in $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$, we can write the data vector $y = (y_0, \ldots, y_{n-1})^T$ as a linear combination of the basis vectors (11):

$$y = a_0\mathbf{c}^0 + \sum_{l=1}^{m}\left[a_{l1}\mathbf{c}^l + a_{l2}\mathbf{s}^l\right] \tag{12}$$

for some coefficients $a_0, a_{l1}, a_{l2}, 1 \le l \le m$. Taking inner products on both sides of the above equation with respect to $\mathbf{c}^j$ (for a fixed $1 \le j \le m$), we get (note all inner products on the right hand side are zero by orthogonality except for $\langle \mathbf{c}^j, \mathbf{c}^j \rangle$):

$$\langle y, \mathbf{c}^j \rangle = a_{j1} \langle \mathbf{c}^j, \mathbf{c}^j \rangle .$$

In Lecture 8, we also saw that $\langle \mathbf{c}^j, \mathbf{c}^j \rangle = n/2$ (see equation (3) in Lecture 8). Thus

$$a_{j1} = \frac{\langle y, \mathbf{c}^j \rangle}{n/2} = \frac{2}{n}\sum_t y_t \cos(2\pi jt/n) = \frac{2}{n}\mathrm{Re}(b_j).$$

One can similarly (by taking inner products on both sides of (12) with respect to $\mathbf{s}^j$) prove that $a_{j2} = -2\mathrm{Im}(b_j)/n$. Finally by taking inner products on both sides of (12) with $\mathbf{c}^0$, we get

$$a_0 = \frac{\langle y, \mathbf{c}^0 \rangle}{\langle \mathbf{c}^0, \mathbf{c}^0 \rangle} = \frac{1}{n}\sum_t y_t = \frac{b_0}{n}.$$

Plugging in these values of a_0, a_{j1}, a_{j2} in (12), we obtain (5). This completes the proof of (5).

8 The case of even n

We assumed that n is odd (and $m = (n-1)/2$). If n is even, then $1/2$ becomes a Fourier frequency and $b_{n/2}$ becomes real (because $\sin(\pi t) = 0$ for all t). In this case, we can simply avoid working with $1/2$ by taking $m = (n-2)/2$ and using the model:

$$\mathrm{Re}(b_j), \mathrm{Im}(b_j) \overset{\text{i.i.d}}{\sim} N(0, \gamma_j^2) \qquad \text{for } j = 1, \ldots, m.$$

This will be equivalent to (8). Basically everything will stay the same as before (only difference is that $m = (n-2)/2$). Here we are essentially forcing $\gamma_{n/2} = 0$. One can try to also try to estimate $\gamma_{n/2}$ using $b_{n/2} \sim N(0, \gamma_{n/2}^2)$ but this approach is slightly more complicated.